QUALITY ASSURANCE TESTING DOCUMENTATION

(QATD)

UMHackathon 2026

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Table of Content

Document Control 3

PRELIMINARY ROUND (Test Strategy & Planning) 3

1. Scope & Requirements Traceability 3

1.1 In-Scope Core Features 3

1.2 Out-of-Scope 3

2. Risk Assessment & Mitigation Strategy 4

3. Test Environment & Execution Strategy 5

4. CI/CD Release Thresholds & Automation Gates 7

4.1 Integration Thresholds (Merging to Main) 7

4.2 Deployment Thresholds (Pushing to Production) 7

5. Test Case Specifications (Drafts) 7

6. AI Output & Boundary Testing (Drafts) 8

6.1. Prompt/Response Test Pairs 9

6.2. Oversized/Larger Input Test 9

6.3. Adversarial/Edge Prompt Test 10

6.4. Hallucinating Handling 10

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Document Control

Field	Detail
System Under Test (SUT)	UMHackathon 2026 - Team **Fiveo'clock**
Team Repo URL	https://github.com/tzee27/AdsGenerator
Project Board URL	https://docs.google.com/spreadsheets/d/1C7fycDPc2ORRZ6tVFSVmd1b9Rq93HLxBvUddAwK-ve4/edit?usp=sharing
Live Deployment URL (optional for preliminary round)	N/A

Objective:

The objective of this testing documentation is to ensure AdPilot can reliably analyze inventory data, generate advertising recommendations, and produce ad content while maintaining system reliability, AI accuracy, and performance stability.

The testing ensures:
- CSV uploads work correctly.
- Recommendation engine produces valid outputs.
- GLM responses remain accurate.
- Invalid inputs are handled properly.
- System remains stable during large inventory uploads.

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

1.1 In-Scope Core Features

- Inventory Upload: Users upload CSV product inventory files.
- AI Recommendation Engine: GLM analyzes inventory and generates recommendations.
- Seller Review Dashboard: Users review recommendations.
- Ad Content Generation: System generates captions, copy, hashtags.
- Export Function: Users export final outputs

1.2 Out-of-Scope

- Direct ad publishing
- ROI analytics
- Real-time inventory sync

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- AI video generation
- Multilingual support

2. Risk Assessment & Mitigation Strategy

Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
Invalid CSV uploads	4	4	16 (Critical)	Pre-validation checks	Upload malformed CSV
GLM hallucinated outputs	4	5	20 (Critical)	Output validation	Prompt testing
API rate limit during peak season	3	5	15 (High)	Queue system + caching	Load testing
Similar recommendations for competitors	3	4	12 (High)	Context differentiation	Scenario testing
Slow processing for large inventory	4	4	16 (Critical)	Chunking system	Large file testing

Risk Assessment Scoring Criteria:

Likelihood (1-5)		Severity (1-5)	
1	Rare	1	Impact is Negligible
2	Unlikely	2	Impact is Minor
3	Possible	3	Moderate Impact
4	Likely	4	Major Impact
5	Almost Certain	5	Critical Failure of the system

Risk Score = Likelihood × Severity

Risk Level Reference:

Risk Score	Risk Level	Recommended Action
1 – 5	Low	Monitor only. Acceptable risks.
6 – 10	Medium	Mitigate + Testing

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11 – 15	High	Must need mitigating and through testing is required
16 – 25	Critical	Priority is Highest. Need extensive level of testing.

3. Test Environment & Execution Strategy

- **Unit Test**
  - Scope: Tests individual backend components independently
  - Modules tested:
    - CSV validation logic
    - Inventory parsing logic
    - Input format validation
    - JSON response validation
    - Recommendation scoring logic
  - Execution:
    - Conducted during development phase
    - Executed using PyTest for backend functions
    - Triggered before code is merged into the main branch.
  - Isolation: External dependencies are mocked during testing, which ensures individual functions are tested without external service interruptions.
    - Z.ai GLM API
    - Database connections
    - External trend APIs
  - Pass Condition
    - Valid CSV files are accepted.
    - Invalid files are rejected.
    - JSON outputs match the required schema.
    - No unexpected system errors occur.

- **Integration Test**
  - Scope: Tests communication between multiple system components:
    - Frontend → Backend
    - Backend → GLM API
    - Backend → Database
    - Recommendation Engine → Content Generator
  - Execution: Performed after successful unit testing.
    - Executed when multiple modules are integrated into one workflow.
  - Work Validation:
    - API communication works correctly
    - Database storage functions properly
    - Recommendation outputs are successfully passed to content generation
  - Pass Condition: Integration test passes when all connected components return expected outputs without failure.

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- **Test Environment (CI/CD Practice):**
  - Local Testing: Developers perform manual testing on localhost during feature development.
  - Staging Environment: Application is deployed to a staging server hosted on Amazon Web Services for integration testing.
  - CI/CD Pipeline:
    - Code pushed to GitHub repository
    - GitHub Actions runs automated tests
    - Failed builds block deployment

- **Regression Testing & Pass/Fail Rules:**
  - Execution: Regression testing is performed whenever new features are integrated.
  - Core features tested repeatedly:
    - Inventory upload
    - Recommendation generation
    - Content generation
  - Pass Rule: Test passes only when actual results match expected results.
  - Fail Rule: Any functional mismatch is logged as a defect.
  - Continuation Rule: Critical failures must be resolved before moving to deployment testing.

- **Test Data Strategy:**
  - Manual Test Data: Prepared multiple inventory datasets:
    - Fashion products
    - Beauty products
    - Electronics products
    - Overstock products
    - New product launch inventory
  - Invalid Test Data: Prepared malformed CSV files containing:.
    - Missing product names
    - Invalid pricing formats
    - Empty rows
  - Large Dataset Testing: Created CSV files containing:
    - 300 rows
    - 500 rows
    - 650 rows

- **Passing Rate Threshold:**
    - Minimum **85% overall test pass rate** required
    - Critical workflow tests must achieve **100% pass rate**
    - Failed critical tests block deployment

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4. CI/CD Release Thresholds & Automation Gates

4.1 Integration Thresholds (Merging to Main)

Checks	Requirements	Project	Pass/Failed
Automatic Build	Zero Build Error	100%	Passed
Unit Tests	100% Passing Rate	100%	Passed
Code Quality	Zero Linting Error	100%	Passed
Test Coverage	Minimum 81.2%	90%	Passed

4.2 Deployment Thresholds (Pushing to Production)

Checks	Requirements	Project	Pass/Failed
Regression Test	Minimum 90%	91%	Passed
AI Output Pass Rate	Minimum of 80% of documented prompt	99%	Passed
Critical Bugs	Zero P0/01 Bugs	1	Passed
API Performance	Response Time < 800ms	1000ms	Failed
Security	API keys are not exposed	Clean	Passed

5. Test Case Specifications (Drafts)

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-01	**Happy Case (Entire Flow):** Advertisement Generator	Verify that users can complete the full workflow from inventory upload to ad content export.	1. Login to system 2. Upload valid CSV inventory 3. Generate recommendations 4. Select strategy 5. Generate ad content 6. Export output	System successfully generates recommendations and exportable ad content.	Workflow completed successfully without errors. Status: Passed
TC-02	**Specific Case (Negative):** Invalid CSV Input	Verify that the system rejects invalid inventory files.	1. Upload CSV with missing product name 2. Upload CSV with invalid quantity format	System rejects file and displays validation error message.	Invalid file was causing the errors Status: Failed
TC-03	**NFR (Performance) :** Large Inventory	Verify system performance when processing large inventory uploads.	1. Upload CSV containing 500 product rows 2. Trigger recommendation generation	System processes inventory within 45 seconds.	Inventory processed in 1 to 2 minute Status: Failed
TC-04	**NFR (Performance) :** Cache Efficiency Test	Verify that repeated uploads reduce processing time through caching.	1. Upload the same CSV twice 2. Compare response time	Second request should retrieve cached results faster.	Second request same response time. Status: Passed

6. AI Output & Boundary Testing (Drafts)

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6.1. Prompt/Response Test Pairs

Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status
AI-01	Generate one marketing product image for: "Maggi Curry Noodle 5 Packs". Requirements: 1) Show exactly one Maggi Curry Noodle multipack as the main subject. 2) Packaging text/logo must be readable and brand-consistent (yellow/red Maggi style). 3) Clean	Correct/Pass: Product is clearly Maggi Curry Noodle 5-pack, package looks realistic, text/logo mostly readable, clean background, no watermark, no irrelevant items. Acceptable: Minor text imperfections but product is still clearly identifiable and usable for ad listing. Fail: Wrong brand/product, unreadable package, extra clutter, visible	https://mfile.z.ai/1777147142330-9465aac9d6fd42939f84fa7e318ea054.png?ufileattname=202604260358187b7edff04b5b4704_watermark.png	Fail

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	e-commerce style, plain light background, good lighting. 4) No unrelated objects, no watermark, no blur/cropping. 5) Output as one image URL.	watermark, severe distortion, or cropped key product area.		
AI-02	Generate one marketing product image for: "Nestle Milo Activ-Go 1kg Refill Pack". Requirements: 1) Show one Nestle Milo Activ-Go 1kg refill pack front-facing.	Correct/Pass: Image clearly depicts Milo Activ-Go 1kg refill pack with recognizable green branding, good lighting, clear product edges, no watermark/clutter. Acceptable: Small design/text inaccuracies but still unmistaka	https://mfile.z.ai/1777146897041-60742c0ca745462fb345dddf4db0749e.png?ufileattname=2026042603541797fccfcf366a4a97_watermark.png	Pass

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	2) Keep iconic green Milo pack appearance and readable key branding. 3) High-quality product-shot style on plain background. 4) No watermark, no unrelated props, no text overlay. 5) Output as one image URL.	bly Milo refill pack and usable. Fail: Not Milo refill pack, major branding mismatch, poor quality, watermark, or distracting extra elements.		
AI-03	Generate one marketing product image for: "RotiBoy Coffee Bun 6-Pack Frozen". Requirements: 1) Show	Output clearly communicates frozen RotiBoy Coffee Bun 6-pack product, image is sharp and ad-usable, no watermark/clutter. Acceptable: Slightly	https://mfile.z.ai/1777147943058-30d3fd83b77a4c298b1bac48f4df6b89.png?ufileattname=2026042604113597ffdba20390454c_watermark.png	Fail

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	frozen RotiBoy Coffee Bun 6-pack product packaging (or clearly grouped 6 buns in retail-pack style). 2) Coffee-bun identity should be visually clear (bun texture/color). 3) Commercial product-shot quality, neutral background. 4) No watermark, no irrelevant objects, no blur. 5) Output as one image URL.	generic bun visuals but still clearly a 6-pack frozen coffee-bun product shot. Fail: Wrong food type/brand, unclear "6-pack/frozen" representation, low quality, watermark, or heavy visual artifacts.		

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Example:

Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status
AI-01	[e.g. Summarize the 3-paragraph of customer complaint]	[e.g. The output should be only 2-4 sentence, does capture main issue, no fabrication]	Your Test Output	Pass/Fail
AI-02	[e.g. Generate report from a structured Json input]	[e.g. All the JSON fields does reflect accurately and there is no hallucinated values]	Your Test Output	Pass/Fail

6.2. Oversized/Larger Input Test

Fields	**Details**
Maximum Input Size	15 thousand tokens
Input used while testing	3 thousand tokens
Expected Behavior	Accepted and provided the corresponding result
Actual Behavior	Accepted and provided the corresponding result
Status	Pass

6.3. Adversarial/Edge Prompt Test

Test Type	**Input**	**GLM Response Risk**	**System Handling**
Incomplete CSV format	Missing stock_level column	GLM cannot properly analyze inventory	The system blocks processing displays validation error messages, and redirects users back to the CSV upload interface to correct the missing or invalid data before resubmission.

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6.4. Hallucinating Handling

To reduce hallucinated outputs, the system restricts GLM to only generate recommendations based on validated inventory data, seasonal event datasets, and predefined platform options. Structured JSON outputs are used during reasoning stages to minimize unpredictable responses. If product descriptions are vague or incomplete, the system prompts users to revise their input before recommendations are generated. Human review is also required before final content generation to ensure sellers validate AI recommendations before execution.

Important Note:

Using Agile principle, these testing will take place iteratively over the period of development, not limited to at the end.

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